Environmental Health Surveillance Manual

Pima County Health Department, Environmental Health Division

Contents

1 Purpose and Scope **3**

2 Definitions **4**

3 Roles and Responsibilities **5**

4 Routine Environmental Health Surveillance Data Sources **6**

5 Program Planning Assumptions **7**

6 Operational Record Review **8**

7 School Outdoor Activity Surveillance Procedures **9**

8 Data Quality and Completeness **10**

9 Analysis and Reporting Standards **11**

10 Privacy and Confidentiality **12**

11 Records Retention **13**

12 Training **14**

Document control 15

Revision history 15

1 Purpose and Scope

This manual states how the Environmental Health Division of the Pima County Health Department conducts environmental public health surveillance and administers the school outdoor activity surveillance program. It compiles the former Environmental Health Surveillance Standard Operating Procedure and the School Outdoor Activity Surveillance Codebook into one controlled document, so that division staff and school reporting contacts work from a single set of definitions and procedures.

The manual applies to all division personnel who collect, review, analyse or report environmental health data, and to the school site staff who submit outdoor activity reports to the division. It covers the county's routine surveillance data sources, the annual program planning cycle, the review of operational records, and the standards that published analyses follow. It does not govern clinical care, individual patient management or enforcement actions, which are handled under separate division policies.

Compliance with this manual is a condition of the division's data-sharing agreements with participating school districts and reporting facilities.

2 Definitions

The following definitions apply throughout this manual and in every report the division publishes.

- **Air Quality Index (AQI).** The index the U.S. Environmental Protection Agency uses to communicate daily air quality, divided into six colour-coded categories that correspond to levels of health concern.
- **Daily AQI value.** The AQI value reported for a county for a calendar day, together with the category, the defining parameter and the monitoring site that defined it.
- **Defining parameter.** The pollutant whose sub-index set the day's AQI value, as recorded in the daily county data the division obtains from EPA AirData.
- **Syndromic surveillance.** The ongoing collection of emergency-department visit records classified by presenting complaint, used to monitor community health conditions without waiting for definitive laboratory results.
- **Asthma-specific emergency-department visit.** An emergency-department visit record classified by the syndromic surveillance registry under the asthma case definition in the registry's classifier version noted on each extract.
- **Outdoor activity.** Any scheduled activity of a school site that takes place outdoors, including recess, physical education classes, athletic practices, scheduled athletic events and field trips.
- **School site.** A reporting location within a school district that submits outdoor activity reports to the division under section 7.
- **County cancellation day.** For county-level analyses, a cancellation day is a calendar date on which at least one school in the county cancelled outdoor activities; multiple school-site records for the same date are one county cancellation day.

3 Roles and Responsibilities

Division Director, Environmental Health. Approves the annual surveillance plan, signs off revisions of this manual and receives the operational record reviews prepared under section 6.

Lead Biostatistician. Designs the division's analyses, prepares the analytic datasets used in published reports, documents the methods and assumptions behind each estimate, and answers for the statistical content of deliverables sent outside the division.

Surveillance Analyst. Maintains the routine data extracts described in section 4, keeps the extract log current and prepares the data quality checks described in section 8.

School Liaison. Maintains the roster of school site reporters, trains new reporting contacts and is the first point of contact for questions about the outdoor activity log.

School site reporters. Submit outdoor activity reports as described in section 7, and tell the School Liaison when a reporting contact changes.

4 Routine Environmental Health Surveillance Data Sources

The division maintains six routine data sources for environmental health surveillance. Each source's refresh cadence and owning role are recorded in the extract log kept by the Surveillance Analyst.

- **EPA AirData daily AQI by county.** Daily county-level AQI records obtained from the U.S. Environmental Protection Agency's AirData system. The division's working copy of this file is prepared by the analytics team and carries the fields the EPA publishes: state and county names, state and county codes, date, AQI value, category, defining parameter, defining site and the number of sites reporting.

- **EPA AirData annual AQI by county.** The annual county summary published by the EPA, with the days in each AQI category, the maximum, median and ninetieth percentile AQI values and the count of days by defining parameter.

- **Census county reference files.** The county FIPS reference and the county population estimates published by the U.S. Census Bureau, used to identify counties and to describe the resident population the division serves.

- **AirNow AQI technical assistance document.** The EPA's technical assistance document for the reporting of daily air quality, which defines the AQI categories, their index value ranges and the colour and health message that goes with each category.

- **The department's syndromic surveillance registry.** Emergency-department visit records from reporting facilities, aggregated by the registry into daily counts by county, with the reporting facility completeness figures described in section 8.

- **The school outdoor activity log.** The reports school sites submit under section 7, one record per school site and date.

When a source publishes a correction or a revised edition, the Surveillance Analyst replaces the division's copy, records the change in the extract log and tells the Lead Biostatistician which pending analyses are affected.

5 Program Planning Assumptions

The division's planning documents set the operating parameters the program is budgeted and staffed around: the surveillance seasons, the reporting schedule, the outreach calendar and the exposure classifications used in program planning. A planning assumption is a statement of how the program expects to operate, written before the year's records exist, and it is kept in the planning document that adopted it.

Planning assumptions are working positions, not findings. Planning assumptions used for exposure classification are checked against observed operational records; when they disagree, the analysis reports both the planned threshold and the threshold estimated from observed operations.

The biannual program plan records each assumption still in force, the document that adopted it and the year it was written, so that a reader can tell an assumed parameter from a measured one.

6 Operational Record Review

The Lead Biostatistician conducts operational record reviews on a twice-yearly schedule. Reviews cover the most recent complete calendar year.

A review states, for each assumption examined: the source and date of the planning assumption, the records examined, what those records show, and whether the assumption held. Reviews are filed with the division's planning materials and summarised for the Division Director.

Where an operational record and a planning assumption disagree, the disagreement is reported to the Division Director in writing before the planning meeting at which the assumption may be revised, so that the meeting decides with both positions in front of it.

7 School Outdoor Activity Surveillance Procedures

Participating school sites report to the division on the days the division requests a report. A report names the school site, the date, whether outdoor activity was cancelled, the reported reason and the activities affected, and the person or role who filed it. Sites file one record for their site and date; a district with several participating sites receives one request per site.

The log covers reported outdoor activities year-round, including summer sessions and Saturday athletics, so a report can be requested for any day on which a site has outdoor activities scheduled. Days on which a site has nothing scheduled outdoors generate no record.

The School Liaison sends the season's reporting schedule to site reporters before each school term and again before summer sessions. Reporters file by the end of the reporting day; late reports are added to the log with the date they cover, not the date they arrived.

The division treats the log as the operational record of the school outdoor activity program. Published summaries of the log follow section 9.

8 Data Quality and Completeness

The registry records, for every county and day, the number of reporting facilities that submitted records and the number expected to report. The completeness share is the reports received divided by the reports expected.

The Surveillance Analyst reviews facility completeness each month. Facilities that miss a reporting cycle are contacted by the School Liaison's counterpart in syndromic surveillance; facilities that miss two consecutive cycles are escalated to the division's data-sharing coordinator.

An extract's version identifier names the registry classifier version it was produced under. The analyst records any change of classifier version in the extract log, because a change in the case definition can move a count from one syndrome class to another.

9 Analysis and Reporting Standards

Every analysis the division publishes states the data sources it used, the calendar period it covers and the definitions of every measure it reports. An analysis that merges sources keeps the merged dataset, so a reviewer can retrace the work.

A published rate names its numerator, its denominator and the period it covers, and states whether the denominator is a population, a count of days or a count of records. A published count states the records it was taken over.

Annual Air Quality Index summary values for a county evaluation are drawn from the single EPA Annual Summary row for the county named in the evaluation and for the calendar year the evaluation examines.

Model-based estimates are reported with their uncertainty. An analysis that fits a statistical model states the model form, the outcome and exposure definitions, the estimation sample and the diagnostics the analyst ran, in enough detail for a qualified reader to judge the fit.

Reports carry the division's document control block: title, author, date, document identifier and revision, so that later readers can tell which edition of an analysis they hold.

10 Privacy and Confidentiality

The surveillance registry holds health information protected by federal and state law. Analyses are conducted on aggregates only; a published table or figure does not identify a person, a household or an individual facility's patients.

Small cell counts are suppressed in any output released outside the division. Requests for surveillance data from outside the department are referred to the division's privacy officer and answered under the department's public records procedure.

Staff with registry access complete privacy training under section 12 before access is granted, and access is reviewed annually.

11 Records Retention

Routine surveillance extracts, the extract log and the school outdoor activity log are retained for seven years from the end of the calendar year they cover, then destroyed under the department's records schedule.

Published reports, operational record reviews and the planning materials of the biannual program plan are retained permanently. Analytic datasets behind a published report are retained with the report they support.

Destruction of records is logged by the destroying staff member and countersigned by the division's records coordinator.

12 Training

New division staff read this manual and complete the department's data governance training before working with surveillance data. School site reporters receive the reporting guide and a short briefing from the School Liaison before their first reporting season.

The division runs an annual refresher covering changes to this manual, the current reporting schedule and the data quality procedures of section 8. Attendance is recorded in the department's training system.

A school site that repeatedly files incomplete reports receives a follow-up visit from the School Liaison before the next reporting season.

Document control

Field	Value
Document identifier	PCHD-EH-SOP-014
Title	Environmental Health Surveillance Manual
Owner	Environmental Health Division, Pima County Health Department
Revision	Rev 2026-07
Effective date	2026-07-15
Review due	2027-07-15

Revision history

Revision	Date	Summary of change	Prepared by
Rev 2024-02	2024-03-11	Initial Environmental Health Surveillance SOP	Nikhil Sharma
Rev 2025-01	2025-01-20	Added data quality and completeness section	Sophia Rogers
Rev 2026-03	2026-03-02	School Outdoor Activity Surveillance Codebook first issued	Andrea Long
Rev 2026-07	2026-07-15	SOP and Codebook compiled into one manual; definitions aligned	Nikhil Sharma